ENERGY & GRID HARDENING · INNOVATELA DUAL-USE HACKATHON · USC, OCTOBER 2026

POWERUP

A startup's disaster layer for its fleet of Base batteries: **PowerUP plans** each home's backup before a wildfire shutoff, **Interlock guards** every order, and neighbors share one **island** behind SCE's switch.

6,874
customers on six SCE circuits lose power, Oct 9 to 18, 2026 (real)

Oct 15
the Bryson Fire starts during the shutoff (real)

The pilot: 456 homes behind SCE switch RAR8848 on the Cuthbert circuit, 9 km from the burn.

16 shutoffs since 2019, 435 hours dark (real SCE records): once at RAR8848, 15 times the whole circuit. Upstream, a 66 kV line into Latigo runs about 650 m inside the Bryson perimeter.

PowerUP plans, over days

OCT 9, 3 AM
WATCH
First wind warning. Medical Baseline homes fill to 100%.

OCT 12
ASK
SCE names Cuthbert. Owners are asked to pre-fill; a 50% floor for the rest.

OCT 14
FILL
The shutoff is sure. Every battery that said yes fills, and nothing is sold.

OCT 15, 4 PM
HOLD
Power off. Each home runs on a budget: 8, 16 or 32 kWh a day.

OCT 17, 10 PM
HAND
Winds stop. PowerUP proposes the restart; SCE's shutoff officer approves.

Interlock guards, in seconds

Every order, from any sender, passes eight checks (authentication, replay protection, each home's backup reserve, hosting capacity and more) and gets the gentlest safe rung: run, clamp, canary at 1% then 10%, hold, or hand to a person.

It can only shrink, slow or stop an order, never create one. PowerUP's own orders pass through it too, and the reserve PowerUP plans for each home becomes the floor Interlock keeps against every sender.

A SILENT BATTERY IS NEVER COUNTED AS POWER

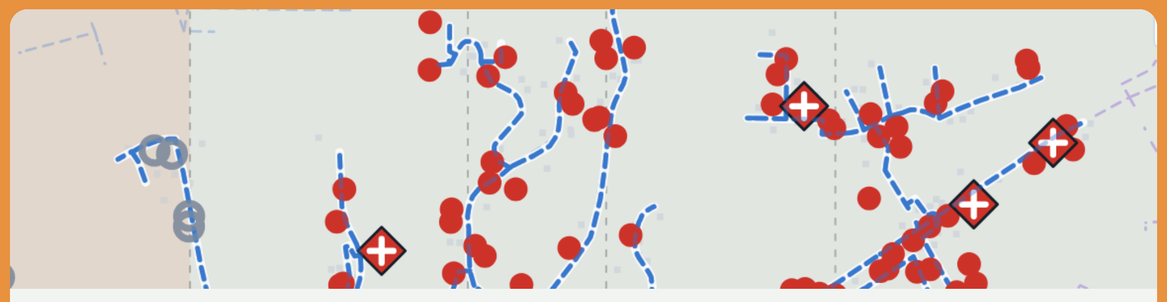
TRIPPED
Its whole circuit went quiet together.

NOT TALKING
Its cell tower or internet went quiet. It follows its plan alone, over mesh to Starlink.

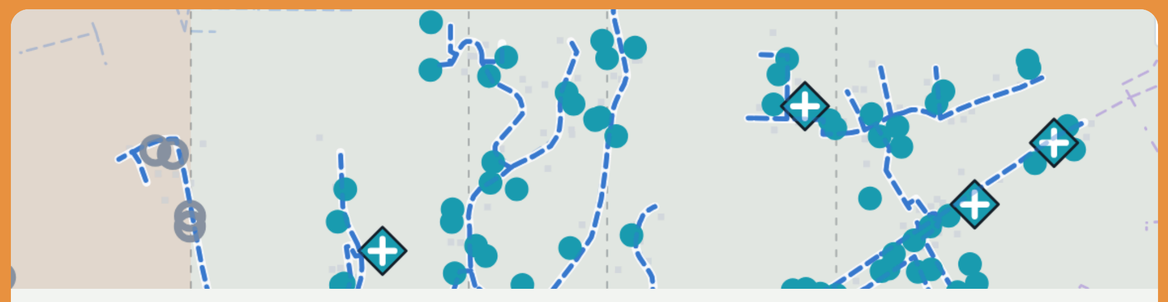
BROKEN
Quiet alone, after a fault code: to a person.

UNSURE
Anything else: ask the owner first.

Same shutoff, six failures: guard off vs guard on



Guard off, a day into the shutoff: batteries empty



Guard on, same hour: every battery home has power

Battery homes that go dark
189 → 0

Medical Baseline homes that go dark
16 → 0

Fleet charge when the power goes off
9% → 99%

Battery-hours onto a dead line
567 → 0

Overload at the restart, Cuthbert
3.4 MW → 0

Home-hours kept powered
945 → 11,596

Simulated failures on the real timeline: a stolen key, a replayed order, a sell-down three hours before the shutoff, export onto the dead line, a recharge rush and SCE's own emergency event. 189 battery homes (160 free + 29 their own, assumed share). PowerUP's own orders held: 0 of 5. Reproduce: **node pilot/app/cli.mjs**.

THE PILOT BUSINESS

160 free Base Core batteries in volunteer homes, Medical Baseline first, then solar roofs. When SCE opens RAR8848, the 456 homes run as one island on them and the solar. They carry all 16 real shutoffs (15 of 16 with smoke). Assumed: SCE can island the section, and our batteries are grid-forming.

$9.99 a month buys emergency power for a home with no battery (226 homes, assumed 90% opt-in). **$1.20 per kWh** in a shutoff for everyone on the island, twice SCE's top rate (assumed), billed by SCE; sellers get $1.00.

FUNDING DOE SBIR (Office of Electricity, CESER) first, DOE GRIP with SCE to scale, FEMA HMGP after a disaster. Built: an installation pack for Naval Base Ventura County.

GLOBAL Australia's bushfire shutoffs and life-support register; South Africa's load shedding, with clinics in the medical role. Built: region policy packs.

EVERY DAY: INCOME PER BATTERY, PER YEAR

LINE	LABEL	$ / YEAR
Trading, CAISO prices for SCE	real prices	334
Capacity payments	assumed	941
Emergency grid events	assumed count	627
Subscriptions	assumed opt-in	169
Shutoff power sales	assumed rates	90
Holding full from SCE's notice	real prices	−28

Fleet about $2.24M after the 30% tax credit (installed cost assumed), about $341,000 a year. Payback 6.6 years with all income, 25 without capacity and event pay, 46 on trading alone.

RUN THE REPLAY YOURSELF

mrbese.github.io/PowerUp/

Code: github.com/mrbese/PowerUp · Every number labelled real, assumed or simulated.

RESPONSIBLE USE

Misuse: locking owners out of their batteries, a single point of attack, or pricing power up mid-crisis. Limits: Interlock never goes below a home's own backup, PowerUP sees a priority tier not a medical record, a person approves every restart and island, and the price is fixed in advance.